Introduction to Kolmogorov-Arnold Networks

Theoretical background, current approaches, potential next steps

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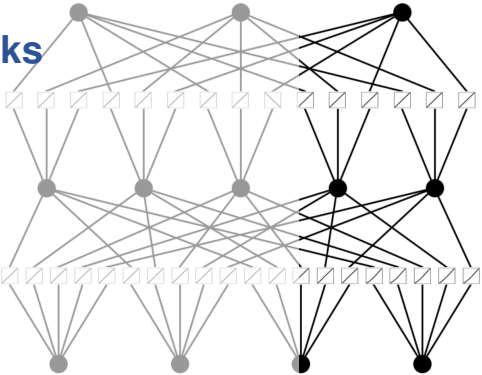

DATA ENGINEERING GROUP

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Motivational Slide

- MLPs are inspired by the Universal Approximation Theorem
  - Leshno et al. (1993): *A standard multilayer FNN can approximate any continuous function to any degree of accuracy iff the network's activation functions are not polynomial*
- Can we be inspired by K-A Representation Theorem to define a new kind of trainable network?
  - Kolmogorov (1956, 1957) and Arnold (1957): *Every continuous multi-variable function can be expressed using composition and addition of uni-variate functions*
- Learning uni-variate functions is more efficient, the results directly visualizable and interpretable

Outline

- Part 1, Kolmogorov-Arnold Representation Theorem: Historical context; proof sketch; thoughts on the theorem's impact
- Part 2, Kolmogorov Arnold Networks: Definition; training algorithm
- Part 3, Why Kolmogorov Arnold Networks? Motivation; expected applications
- Part 4, Life after KAN: Extensions; DEG current work and plans

Part 1. **Kolmogorov-Arnold Representation Theorem**

Historical Context

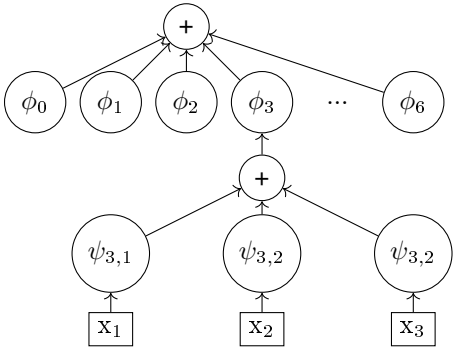
- Every 7th-degree can be algebraically reduced to $\mathrm{x}^7 + \mathrm{ax}^3 + \mathrm{bx}^2 + \mathrm{cx} + 1 = 0$ (Tschirnhaus, 1683)
  - Solving this means finding the three-variable function that maps $\mathrm{a}, \mathrm{b}, \mathrm{c}$ to x
- Hilbert's 13th (1900, 1902): Can this function be expressed as the composition of a finite number of two-variable functions?
- Kolmogorov (1956): Every continuous multi-variable function can be expressed as the composition of a finite number of three-variable functions
- Arnold (1957): Only two-variable functions are really needed
- Kolmogorov (1957): Actually, the only two-variable function needed is addition
  - A generalization of Hilbert's question, except that the univariate functions are not necessarily algebraic

Kolmogorov-Arnold Representation Theorem

- Every multi-variate continuous function f can be written using only univariate functions and addition:

$$\mathrm{f}\left(\mathrm{x}_{\mathrm{i}=1..\mathrm{n}}\right) = \sum_{\mathrm{q}=0}^{2\mathrm{n}} \phi_{\mathrm{q}} \left(\sum_{\mathrm{p}=1}^{\mathrm{n}} \psi_{\mathrm{q,p}}(\mathrm{x}_{\mathrm{p}}) \right)$$

- In other words, addition is the only real multi-variate function, everything else is syntactic sugar

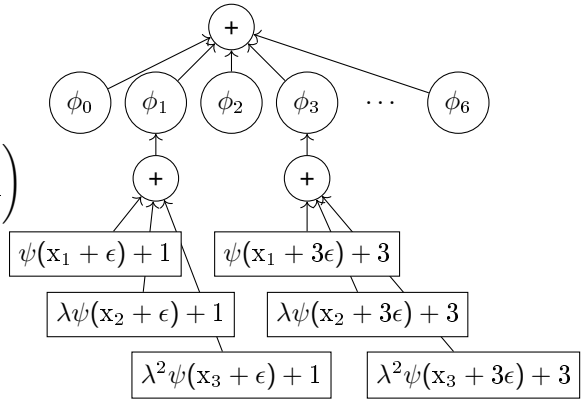


Sprecher Formulation

Sprecher (1972) gives an alternative formulation with a single function at the lower layer:

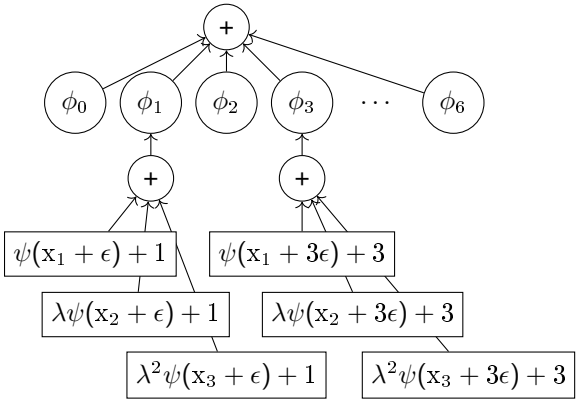
$$\mathrm{f}(\mathrm{x}_{\mathrm{i}=1..\mathrm{n}}) = \sum_{\mathrm{q}=0}^{2\mathrm{n}} \phi_{\mathrm{q}} \left(\sum_{\mathrm{p}=1}^{\mathrm{n}} \lambda^{\mathrm{p}-1} \psi(\mathrm{x}_{\mathrm{p}} + \mathrm{q} \cdot \epsilon) + \mathrm{q} \right)$$

where ϕ_{q} are continuous, ψ is monotonically increasing and Lipschitz-1 continuous, λ and ψ depend only on n and not on f, and ϵ is any non-zero constant



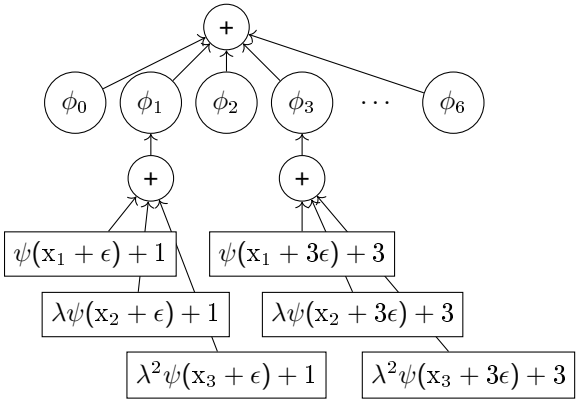
Some Intuitions

- Sprecher uses powers of λ but other non-linear sequences also work
- Feels like multiple values in the unit interval are 'packed' into a single value in $\mathcal{R}$
- Think how addition of powers of 10 packs single-digit numbers into multiple-digit numbers in a reversible way, 42 can only be the sum of the powers of 4 and 2.



Some Intuitions

- ψ has many 'intervals' and translating by $\mathrm{q} \cdot \epsilon$ lands $\mathrm{x_i}$ into the right interval
  - hacks the multiple ψ_i functions into one that behaves differently in each interval
- The lower level 'packs' the effect of f on $\mathrm{x_i}$ for different values of the other inputs $\mathrm{x_{j \neq i}}$
- The ϕ functions can now 'decode' the sums into the original values and calculate the result.



Proof Sketch

The actual proof is between E^{n} and $\mathrm{E}^{\mathrm{n}-1}$, which gives inductively that a function in E^{n} can be represented with functions in E.
To allow for nice figures, we will show that for any $\mathrm{f} : \mathrm{E}^2 \to \mathcal{R}$ there are $\phi_{0..4} : \mathcal{R} \to \mathcal{R}$ and $\psi_{0..4,1..2} : \mathrm{E} \to \mathcal{R}$ such that:

$$\mathrm{f}(\mathrm{x}_1, \mathrm{x}_2) = \sum_{\mathrm{q}=0}^{4} \phi_{\mathrm{q}} \left(\psi_{\mathrm{q},1} \left(\mathrm{x}_1 \right) + \psi_{\mathrm{q},2} \left(\mathrm{x}_2 \right) \right)$$

The proof proceeds in three steps:

- Break the input range into non-overlapping intervals with gaps
- Construct the inner functions ψ
- Construct the outer functions ϕ

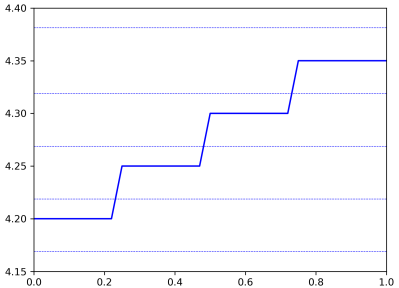
Proof Sketch Step 2: Construct the inner functions

For given k non-overlapping intervals, it is always possible to construct a series of constants λ_i such that:

$$\lambda_\mathrm{i} < \lambda_{\mathrm{i}+1} \leq \lambda_\mathrm{i} + \frac{1}{2^\mathrm{k}}$$

and a function ψ_1 mapping inputs from the intervals of the inputs to the corresponding intervals of the λ_i series.

Any continuous function will do, but let's say it's a step function with linear connections across gaps, so that it's continuous and monotonic increasing.



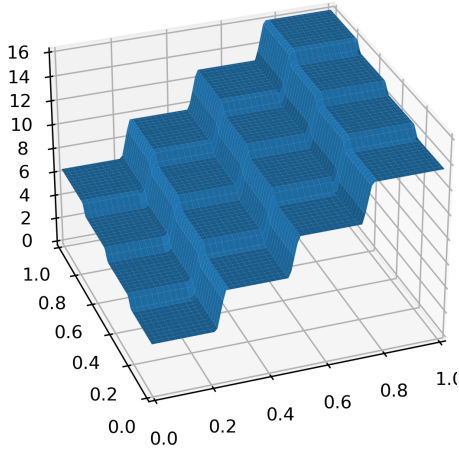
For $\mathrm{k} = 4$, four non-overlapping intervals and the corresponding 'lanes' defined by the λ series.

Proof Sketch Step 2: Construct the inner functions

Construct a similar series for the second dimension; Then... Add.

There are many solutions, but Sprecher's (1972) formulation gives us something we can visualize:

A grid of non-overlapping 'platforms', linearly connected at the gaps, all having different z, so that each z maps to exactly one 'platform' or to a gap.



Proof Sketch Step 2: Construct the inner functions

We repeat all the above for to create $2\mathrm{n}+1$ different systems of 'platforms', one for each upper function.

Step 1 constructed the non-overlapping intervals and *also*

- includes the proof that these intervals can always be constructed so that every point in E^2 is covered by at least one non-gap area of one of these systems of 'platforms'
- *Intuition:* Do not align the gaps

So:

- For each of the $2\mathrm{n}+1$ outer functions, we have created the inner functions that map from *some* values of $\psi_1(\mathrm{x}_1)+\psi_2(\mathrm{x}_2)$ to a platform
- Some other values might be over a gap.
- *But:* There is no value that is over a gap in *all* systems of platforms

Proof Sketch Step 3: Construct the outer functions

All the pieces are now in place:

- We have an 'index' from $\psi_1(\mathrm{x}_1) + \psi_2(\mathrm{x}_2)$ to squares restricting the possible values of $\mathrm{x}_1, \mathrm{x}_2$. The size of the squares gets smaller as k gets larger.
- We can construct a function f_k that simply 'memorizes' how to approximate the value of the original f:
  - all pairs in the unit square are covered
  - f_k knows the neighbourhood of the $\mathrm{x}_1, \mathrm{x}_2$ values despite the fact that it only sees sums
  - f_k can estimate f with some error bounded by an expression denominated by k
- As the number of intervals k increases, the error decreases. At the limit, there is no error.

Unlike the ψ functions, we have no intuitions whatsoever about what the ϕ functions look like or how to construct them.

Some Thoughts and Intuitions

- The theorem premises that x_i are in the unit interval—this is needed for some of the constructions in the proof and is not a real restriction
- The theorem proves the existence of the ϕ and ψ functions, does not tell you how to construct them
- We know very little about ϕ (continuous) and slightly more (but still little) about ψ (monotonic increasing and Lip-1 continuous)
- It is unlikely that these functions can be defined by a neat-looking closed formula

Some More Thoughts

There are many ways to pack two real numbers into one and then define the function that unpacks the original numbers and applies f. Here is one packing schema:

$$\begin{array}{l} 0.42 \\ 0.31415926\ldots \\ \hline 0.4321040105090206\ldots \end{array}$$

The Kolmogorov-Arnold therom tells us that it is possible to find a **continuous** function that does the same

- 'Continuous' sounds like a positive quality, but many 'pathological' functions are nominally continuous
- Under this light, we consider the claim that addition is the only multi-variate function strictly speaking valid, but over-hyped

Some More Thoughts

Kudos to Kolmogorov and Arnold for:

- **Anticipating modern ML**: As long as it fits the data, it doesn't matter if the model captures the actual structures that generate the data
- **Anticipating modern ML II**: In the executive summary, over-hype the significance of the result
- Anticipating that multiple arguments can be passed as one complex record a couple of years before COBOL

Post-KART

- Hilbert's intuition about not being able to represent more complex functions by algebraically combining simpler functions remains valid:
  - It obviously remains open for algebraic functions
  - Its essense holds for any function, except that the K-A Representation Theorem proves that using the number of arguments to define complexity is wrong
  - Vitushkin (2004) gives a compelling argument
- Hecht-Nielsen (1987) notes the similarity between Sprecher's formulation and NNs and speculates that NN training can be used to construct ϕ, but Girosi & Poggio (1989) point out that we know ϕ are not smooth (Vitushkin & Henkin, 1967) and cannot be learnt.
- Schmidt-Hieber (2021) gives a good overview of the ways the KART has interacted with the ML community.

Part 2. **Kolmogorov-Arnold Networks**

Neural Networks are Function Approximators

- Back to our original motivation: Can we be inspired by KART to define a new kind of trainable network?
  - KART claims that learning a high-dimensional function reduces to learning a polynomial number of 1D functions then, why **inspired**? . . .
  - . . . these 1D functions can be non-smooth (aka *not learnable!*)
  - . . . using only 2-layer nonlinearities and $2\mathrm{n}+1$ terms in the hidden layer is pretty restrictive

- *But*,
  - We can argue that most ML-concerning functions are smooth, and
  - We can just stack ϕ layers and pray

- So, we are no longer using KA(R)T but moving towards a KA(A)T

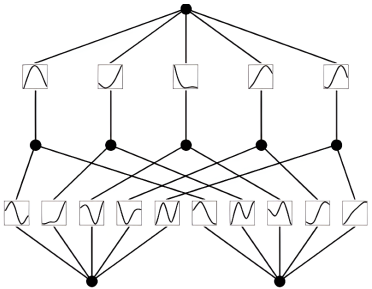
- This is how Kolmogorov-Arnold Networks (KANs) are born (Ziming Liu et al., 2024)

Architecture: High Level

- *Inspiration:* KART

$$\mathrm{f}(\mathrm{x}_1, \ldots, \mathrm{x}_\mathrm{n}) = \sum_{\mathrm{q}=0}^{2\mathrm{n}} \phi_\mathrm{q} \left(\sum_{\mathrm{p}=1}^{\mathrm{n}} \psi_{\mathrm{q},\mathrm{p}}(\mathrm{x}_\mathrm{p}) \right)$$

- We just need to learn the univariate functions ψ and ϕ and perform addition
- ($\rightarrow$) A shallow KAN representing the above equation, with $\mathrm{n} = 2$ input variables x_p, $2\mathrm{n} + 1 = 5$ ϕ_s and $\mathrm{f}(\mathrm{x}_1, \mathrm{x}_2)$ as the output
- Learnable activation functions are positioned on edges and summation is performed on nodes



Architecture: Comparison with MLP

Shallow Formula

- Though KART ensures exact representation and UAT guarantees approximation, relaxing KART's assumptions positions both as function approximators
- UAT lacks guarantees on shallow MLP size, and KART provides no method to compute the relative functions
- MLP uses fixed activations on nodes and learnable weights on edges, while KAN employs fixed learnable functions on edges and summation on nodes

Architecture: Comparison with MLP

Deep Formula

- Shifting to deep formula to avoid UAT's and KART's restrictions
- KAN shifts non-linearity computation to the composition of ϕ_s rather than the weights and activation functions in MLPs
- KAN defines a *KAN layer* with n_{in}-D inputs and n_{out}-D outputs as a matrix of 1D functions, stacking these layers (note how we are drifting further away from KART)
- So, the dimensions of the KAN are *not* restricted to $[n, 2n + 1, 1]$

Architecture: Lower Level

- The learnable univariate functions are parametrized as a **B-spline (basis spline) curves** with learnable coefficients of local B-splines
- Splines are a transformation that maps control points to a smooth curve, aiming to generally follow the shape of these points
- Let us present the B-splines in depth

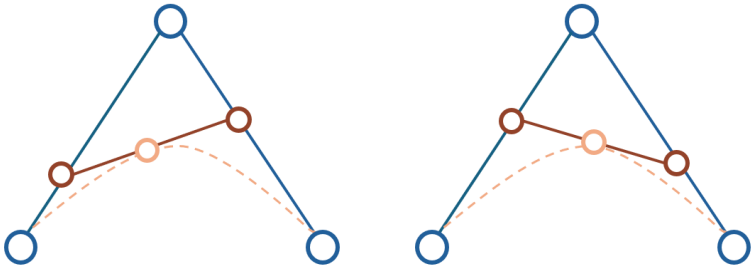
Bézier curves: Linear

To create a path between 2 points P_0 and P_1, we perform **linear interpolation** :

$$\mathrm{P}(\mathrm{t}) = (1 - \mathrm{t})\mathrm{P}_0 + \mathrm{t}\mathrm{P}_1$$

Bézier curves: Quadratic

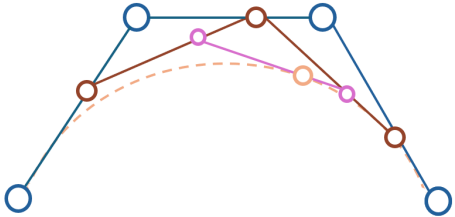
- If we add a third point P_2, then we have two lines connecting them and we can perform linear interpolation to them as well



- This curve is called a **quadratic** Bézier curve

Bézier curves: Cubic and higher

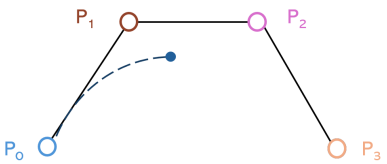
- In the same spirit, we can define the **cubic Bézier curve** (and any other degree, respectively)

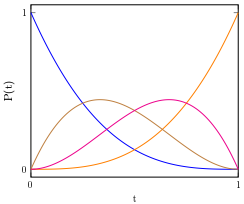


- We call the points **control points**, since the control the curvature

Bézier curves: Algorithms

- This repeated linear interpolation is called the **de Casteljau** algorithm
- A less expensive method for computing Bézier curves involves solving the **Bernstein polynomials** which are the linear interpolation equations expressed wrt the points



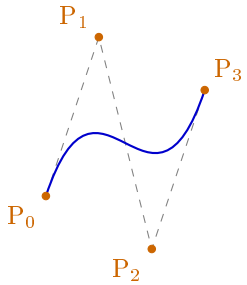


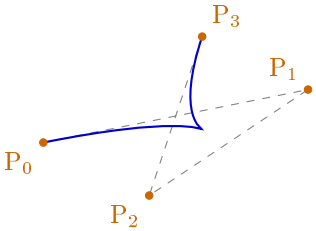
$$\mathrm{P(t)} = \mathrm{P_0} \cdot (-\mathrm{t}^3 + 3\mathrm{t}^2 - 3\mathrm{t} + 1) + \mathrm{P_1} \cdot (3\mathrm{t}^3 - 6\mathrm{t}^2 + 3\mathrm{t}) + \mathrm{P_2} \cdot (-3\mathrm{t}^3 + 3\mathrm{t}^2) + \mathrm{P_3} \cdot (\mathrm{t}^3)$$

- If we plot the factors of these polynomials we get a visualization of the influences of the control points (cf. right Illustration)

Bézier curves: (lack of) local control

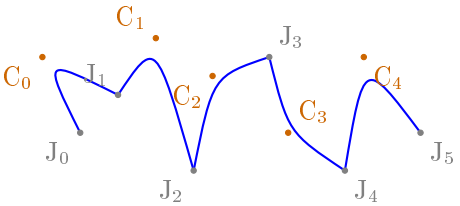
- One main downside of Bézier curves is the lack of local control
- This means that by changing **only one** control point, the curve can change drastically
- Pay attention because we will need this later on!

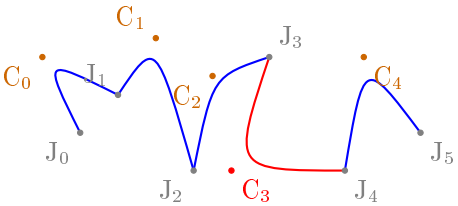




Bézier splines

- How to obtain local control? **Concatenate Bézier curves** → Bézier splines





- Additionally, this alleviates the issue of a high polynomial degree

Bézier splines: Continuity

- In the previous example the curve is not very smooth
- To fix that, we introduce the notion of **continuity**
- Bézier splines can have different levels of continuity at the joint
  - C^0-continuity refers to adjacent Bézier curves ending and starting at the same point
  - C^1-continuity requires C^0-continuity and matching first derivative vectors at the joint
  - C^2-continuity requires C^1-continuity and matching second derivative vectors at the joint
- For **smooth curves**, C^2-continuity is required

B-splines

- B-splines (basis splines) are C^2-continuous Bézier splines
- B-splines of order k are polynomial basis functions for splines of the same order, where, any spline can be represented as a linear combination of B-splines
- Additionally, there exists a unique combination of B-splines for any given spline (de Boor, 1978)
- B-splines are fitting candidates for performing the so-called **spline interpolation**

B-splines: Spline interpolation

- Spline interpolation is used to approximate functions by fitting splines, rather than using a single high-degree polynomial curve
- This process arises naturally from the formal definition of B-splines: it recursively defines each basis from a knot vector (Kaihuai, 1998), which indicates the positions where the splines should be connected
- Given grid G we define a spline S of k^{th} order as:

$$\mathrm{S}_{\mathrm{k,G}}(\mathrm{x}) = \sum_{\mathrm{i}} \mathrm{c_i B_{i,k}}(\mathrm{x}), \text{ where}$$

  $\mathrm{c_i}$ are the coefficients and $\mathrm{B_{i,k}}(\mathrm{x})$ are the corresponding basis functions
  - Liu et al. call grid what is usually called a uniform knot vector and coefficients what are usually called control points
- This process maximizes pre-computed elements (*minimizing possible learnable parameters*)

B-splines and KANs

- In KANs, each activation function is a k^{th} order B-spline, with a G-valued and uniformly spaced grid
- For each activation function what is left to learn is the coefficients of the spline
- Note that KAN also extends the grid on the edges of the spline, so that each grid point is influenced from the same number of polynomials
- Thus, for a KAN of width W and depth D the #training parameters is $\mathbb{O}(\mathrm{W}^2\mathrm{DG})$
- The respective number of parameters for an MLP is $\mathbb{O}(\mathrm{W}^2\mathrm{D})$, but KAN promises convergence with smaller widths

KANs: Implementation details

The authors used a few (questionable) tricks to train the network for optimal performance:

1. *Residual activation functions*:

$$\psi(x) = w_b \cdot \mathrm{silu}(x) + w_s \cdot \mathrm{spline}(x), \text{ where } \mathrm{spline}(x) = \sum_i c_i B_i(x)$$

(One can argue that this introduction of weights does not align with the initial claim)

2. *Initialization scales*: The network's training is sensitive to initialization, so an optimal setup is introduced

Part 3. **Why Kolmogorov-Arnold Networks?**

Grid extension

- For learning a B-spline, more basis functions result in a finer-grained curve
- A very useful feature of KANs is their ability to at first learn a coarse curve for a specific number of grid points and then, at any point, resume the training with more grid points
- The network does not need to be re-trained, but instead they employ a smart idea:

The coefficients of the fine-grained curve can be initialized from the coefficients of the coarse by minimizing the distance between the respective splines (over some distribution of input values)

Fixed activation functions

- KANs also give the option instead of using learnable activation functions, to use **fixed symbolic functions** (polynomials, exponentials etc.)
- This is useful for problems where we have an intuition about their internal functions
- However, we cannot simply set the activation function to be the exact symbolic formula, since its inputs and outputs may have shifts and scalings
- So, in this case, the network learns the parameters of the affine transformation

$$\mathrm{c} \cdot \psi_{\mathrm{fixed}}(\mathrm{a} \cdot \mathrm{x}_{\mathrm{in}} + \mathrm{b}) + \mathrm{d}$$

Interpretability: Motivation

- The main selling point of KANs is their interpretable structure, allowing **retrieval and examination of the learned activation functions**
- But, this is only useful when we a priori know the structure of the KAN and know what we expect to see
- For unknown problems, we stack KAN layers and experiment to find what works
- In this case, only some activation should be meaningful
- Then, how is the visualization considered interpretability?
- The idea is to start from a large enough KAN and train it with sparsity regularization followed by pruning

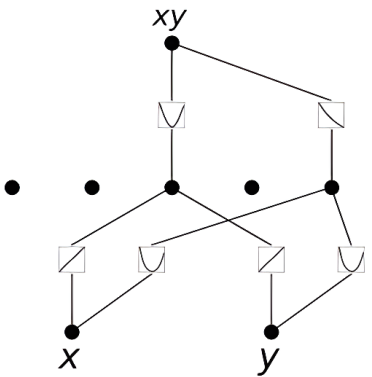
Interpretability: Sparsification

- In MLPs L1 regularization adds the absolute values of the model's weights to the loss function, promoting sparsity by encouraging smaller or zero weights
- Since KANs do not have linear weights, the L1-norm should be defined over the activation functions
- The L1 norm of an activation function (ψ) is its average magnitude over its inputs
- The L1 norm of a KAN layer (ϕ) is the sum of L1 norms of all ψ's
- Actually, L1-norm alone is insufficient, since the goal is to remove less informative or redundant features, not just to shrink weights, so an entropy penalty is introduced (layer-wise)
- So, the total training objective is the prediction loss plus L1 and entropy regularization of all KAN layers

Interpretability: Pruning

- After training with sparsification penalty, we can prune the network to a smaller subnetwork
- The incentive is to throw away the unimportant activation functions
- For each node, its incoming score is defined as the maximum L1 norm of the outputs of the prior KAN layer
- Its outgoing score is defined as the maximum L1 norm of the outputs of the current KAN layer
- A node is considered to be important if **both incoming and outgoing scores are greater than a threshold hyperparameter**

Interpretability: Example



- Function to be learned $\mathrm{f}(\mathrm{x},\mathrm{y}) = \mathrm{xy}$
- KAN computes:

$$(\mathrm{x}+\mathrm{y})^2 - (\mathrm{x}^2+\mathrm{y}^2)$$

- Note that:

$$(\mathrm{x}+\mathrm{y})^2 - (\mathrm{x}^2+\mathrm{y}^2) = 2\mathrm{xy}$$

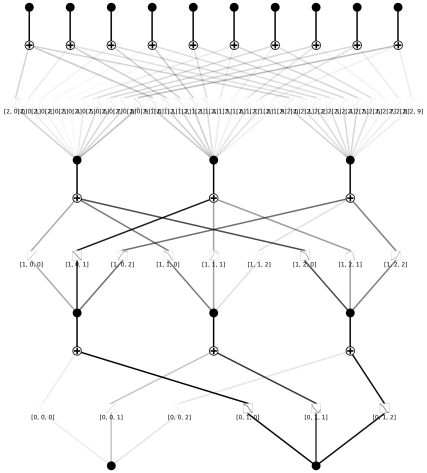
Beating the COD (?)

- The **curse of dimensionality (COD)** refers to the exponential growth in volume associated with adding more dimensions to a space
- UAT tells us that theoretically, a neural network with one hidden layer can approximate any function, but as the dimensionality of the input increases, the number of neurons required grows exponentially
- This leads to a MLP that is very large and potentially computationally intractable
- On the other hand, KAN's approximation ability is more dependent on the number of control points in each activation function and less on the data's dimension
- So, KAN's size will not increase exponentially as the data dimensions increase, but its grid should become more fine-grained
- *However*, we should note that the current (spline) implementation of KANs is not as optimized as the one for MLPs, so in practice the training times for high-dimensional data are comparable

Continual Learning

- Catastrophic forgetting is a problem in MLPs
- When a MLPs is trained on one task and then shifted to being trained on another, the network will soon forget about how to perform the first task
- Do you recall the motivation behind using splines to approximate curves instead of using polynomials? Locality!
- Since spline bases are local, a sample will only affect a few nearby spline coefficients, leaving far-away coefficients intact
- This is desirable assuming faraway regions may have already stored information that we want to preserve
- By contrast, since MLPs usually use global activations any local change may propagate to regions far away

Kind reminder: KAN is yet another function approximator!



Part 4. **Life after KAN**

Alterations

- Different representation theorem

Name	TLDR	Reference
Deep Learning Alternatives of the Kolmogorov Superposition Theorem	Utilizes the Laczkovich Representation Theorem to reduce the number of learned parameters	Guilhoto and Perdikaris (2025)

Alterations

- Different internal representation – with the main objective being efficiency

Name	TLDR	Reference
Chebyshev Polynomial-Based Kolmogorov-Arnold Networks	Activattion functions are modeled with Chebyshev Polynomials	Sidharth et al. (2024)
fKAN: Fractional Kolmogorov-Arnold Networks with trainable Jacobi basis functions	Utilizes fractional-orthogonal Jacobi functions as basis functions	Aghaei (2025)
FastKAN	B-slines are approximated with Gaussian radial basis functions.	Li (2024)
Wavelet Kolmogorov-Arnold Networks (Wav-KAN)	Uses wavelets instead of B-splines	Bozorgasl and Chen (2024)

Faster KANs → sacrifices are made

Extensions

Temporal

Name	TLDR	Reference
seqKAN: Sequence processing with Kolmogorov-Arnold Networks	Replaces the hidden and output layers of an RNN with pure KAN functions, i.e., without trainable weights	Boura and Konstantopoulos (2025)
Temporal Kolmogorov-Arnold Networks (TKAN)	Changes the output of the LSTM cell to $\sigma(\text{KAN}(\mathrm{x}))$	Genet and Inzirillo (2024)
KANsformer	An encoder-decoder architecture that utilizes KANs in the decoder	Xie et. al (2024)

Extensions

Convolutions

Name	TLDR	Reference
Convolutional Kolmogorov-Arnold Networks	Replaces convolutional kernels with KAN kernels	Bodner et al. (2025)

Graph

Name	TLDR	Reference
GKAN: Graph Kolmogorov-Arnold Networks	Performs the aggregation of the graph nodes with KANs	Kiamari et al. (2024)

Quantum

Name	TLDR	Reference
QKAN: Quantum Kolmogorov-Arnold Networks	Exploits quantum linear algebra to apply parameterized activation functions on the edges of the network	Ivashkov et al. (2024)

Applications

Domain/Application	Name	TLDR	Reference
Physics Informed Modeling	Kolmogorov-Arnold Network Ordinary Differential Equations (KAN-ODEs)	Vanilla KAN used as an approximator for a dynamical system in the neural ODE framework	Koenig et al. (2024)
Physics Informed Modeling	Kolmogorov-Arnold Network	Enhance KANs networks with a physics informed loss	Liu et al. (2024)
Human Activity Recognition	iKAN	Incremental Learning paired with KANs; Encoder and frozen KAN-based classifier training	Liu et al. (2024)
Medical Diagnostics	Bayesian Kolmogorov Arnold Networks (BKANs)	Utilize probabilistic splines to better tackle uncertainty	Hassan (2024)
EEG processing	KAN-EEG	Utilization of KANs as a key component for identifying epileptic seizures from pre-recorded EEG signals	Conteras et al. (2025)
Transportation and Mobility	TrafficKAN-GCN	Utilization of a Graph Convolutional-based Kolmogorov-Arnold network for traffic flow optimization	Zhang et al. (2025)
Environment and Energy	Smart home energy prediction framework	Utilization of temporal Kolmogorov-Arnold transformer (TKAT)	Lu et al. (2025)

Possible things to look into

- Try to enforce a ‘separation of concerns’, making more localized changes to account for observed loss
- Experiment with non-BP parameter estimators

Possible things to look into

TUGraz – PML tailored slide.. Just throwing (not so deep) ideas

- **Learning Probabilistic Circuits with Theoretical Constraints from KART?** Structure learning in PCs by introducing KAN-inspired constraints, such as enforcing that the model structure mirrors a composition of univariate functions?

- **Probabilistic Reformulation of KA Superposition?** Can we extend KA to distributions? Can complex joint distributions be decomposed analogously using compositions of low-dimensional distributions?

- **Invertible Probabilistic Circuits via KA Decomposition?** Can we replace standard flow coupling layers with compositions of learned univariate functions and additive operations?

Thank you! Questions?

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Code, bibliography, slides:
`https://github.com/data-eng/kan`